More Than the Sum of Its Parts? Markups and the Role of Establishments

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May 10, 2026

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Abstract

We study distortions to firm production along the intensive margin (how much to produce at each establishment or of each product) and the extensive margin (how many establishments or products to operate). Using data on the universe of Swedish establishments in services industries, we show that 1) firms with larger establishments set higher markups but firms with more establishments do not, and 2) each successive establishment at a firm tends to be smaller relative to its municipality-industry. We interpret these to mean that 1) market power is determined at the establishment-level not the firm-level, and 2) firms' marginal establishments are their smallest. In a model of competition between firms through establishments, we characterize the distortions implied by establishment-level market power: firms inefficiently overvalue production at small vs. large establishments, opening small vs. large establishments, and opening new establishments vs. producing at existing establishments. Calibrated to our Swedish data, it is optimal to tax large high markup firms because they open too many establishments, contrary to findings from models with firm-level market power. Although a high markup indicates that a firm inefficiently produces too little at its existing establishments, it is a poor indicator of the efficiency of the firm's extensive margin.

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1 Introduction

A long theorized problem with large firms is that they have market power, so they produce less than what is socially optimal in order to keep prices high. In a recent class of models supported by empirical evidence, this appears as firm size-dependent markups: within-industries, larger firms set higher markups of price relative to marginal cost.[1] As a consequence, there is a misallocation of production from large firms to small firms. Thus, optimal policy encourages large firms to produce more, for example by subsidizing their sales. Although controversial, this at least suggests that a shift in production toward large firms with high markups—as occurred recently in the United States—may reflect an improvement in allocative efficiency.[2]

This literature mostly treats firm production as one-dimensional.[3] However, in many settings, firms produce along both an extensive margin (how many products to sell or establishments to open) and an intensive margin (how much to sell of each product or at each establishment). For example, expansion by opening new establishments accounts for much of the recent rise of large firms in services industries in the United States.[4] If market power (the demand elasticity) is determined at the establishment-level rather than the firm-level, then firms may produce too little compared to the social optimum at some establishments but too much at others; or they may underproduce at their existing establishments but overproduce by opening too many establishments. What are the implications for misallocation and optimal policy? Do we still want to encourage production at large high markup firms? This question is relevant because it is difficult to design policy that takes into account the full distribution of a firm's products or establishments, which may be hard to observe or even define. Furthermore, implicit and explicit taxes/subsidies that depend on a firm's overall size are a reality of our economic environment.

We argue that a firm's overall size and markup are not useful guides to whether we should encourage it to produce more. A firm's markup indicates whether it produces too much or too little at its existing establishments. But large firms also adjust along their extensive margin by opening/closing establishments. Firms overvalue expanding along the extensive margin relative to the intensive margin, particularly if their marginal establishments are

[1]For example, see Edmond et al. (2023).

[2]See Baqaee and Farhi (2020) for a general analysis. In particular, Weiss (2020) shows that a shift toward intangible capital favors large firms, leading to a rise in industry concentration and welfare.

[3]Notable exceptions are Afrouzi et al. (2023) and Becker et al. (2024), which we discuss below.

[4]See Hsieh and Rossi-Hansberg (2023) and Cao et al. (2022).

small compared to their average establishments, which we estimate to be the case. Indeed, in our calibrated model, it is optimal to tax the largest firms with the highest markups.

We begin with an empirical analysis using data on the universe of Swedish firms and establishments in services industries. In services industries in Sweden, multi-establishment firms are rare, but have enough establishments and relatively large establishments so they earn almost half of all sales. Our first empirical result is that firms with larger establishments tend to set higher markups, but firms with more establishments do not. This suggests that market power is determined at the establishment-level rather than at the firm-level, with larger establishments having more market power. Second, we find that each successive establishment at a firm tends to be smaller, controlling for each establishment's industry and municipality. This indicates that a firm's marginal establishment—the one it will most readily open/close—is its smallest. This has intuitive appeal in the sense that as firms expand, they must work their way down to successively worse establishment opportunities.

We then develop a model of firm competition through establishments, which we study theoretically and quantitatively. Each firm has an initial establishment and faces a firm-specific distribution of additional establishment opportunities, characterized by quality (a demand shifter) and an opening cost. They choose which establishments to open and how much to sell at each. Non-CES Kimball (1995) demand across establishments, without regard to the identity of their firm, implies that larger establishments set higher markups.

Our theoretical result characterizes the inefficient distortions in the market equilibrium implied by these size-dependent establishment markups. All costs compete for a fixed supply of labor, so distortions arise because firms undervalue some uses of labor relative to others compared to a social planner. We show that firms 1) undervalue producing at large establishments relative to producing at small establishments, 2) undervalue opening large establishments relative to opening small establishments, and 3) undervalue producing at existing establishments relative to opening new establishments. By contrast, if market power (demand elasticities) were determined at the firm-level, then larger firms with higher markups would equally undervalue all forms of production—selling more at existing establishments and opening new establishments of any size—compared to smaller firms with lower markups.

Finally, we calibrate a quantitative version of the model to our Swedish data, compute the efficient allocation, and study firm size-dependent policy, which is a tax/subsidy scheme that conditions only on a firm's overall sales. In the calibrated model, firms are characterized by a firm-specific baseline quality, which shifts the quality of all their establishment opportunities. Otherwise, all firms face the same distribution of establishment opportunities in terms of quality and opening costs. Firms with a sufficiently high baseline quality expand beyond their initial establishment and become multi-establishment. Then, a higher baseline quality means they open more establishments, ordered so that each successive establishment has a lower quality, which matches our empirical finding that each successive establishment is smaller. In our main calibration, firms face a linear cost of opening establishments, so all multi-establishment firms have the same sized marginal establishment. Thus, larger firms have larger establishments, higher markups, and more establishments, as in the data.

For firm size-dependent policy, the key mechanism is a tension between the distortions to the intensive and extensive margins of production. Larger firms inefficiently underproduce at their existing establishments because they have larger establishments with higher markups. But they open too many establishments because their marginal establishments are relatively small, which firms overvalue. As a result, the optimal firm size-dependent policy is non-monotonic. It subsidizes production at medium-sized firms, which are single-establishment and respond only by producing more at that establishment. But it taxes small firms as well as the largest firms with the highest markups. If we ignore the extensive margin by taking the set of establishments as given, it is optimal to give the biggest subsidy to the largest firms because they set the highest markups. But this policy backfires badly because these firms respond by inefficiently opening increasingly low quality establishments, leading to lower welfare than without any policy. Ultimately, the extensive margin reduces the maximum gains from firm size-dependent policy by almost 90%, rendering it ineffective.

Expanding through establishments. This paper relates to a recent literature studying firm expansion through opening new establishments. Hsieh and Rossi-Hansberg (2023) and Cao et al. (2022) show that the recent rise in industry concentration in the US services sector was due to the largest firms expanding by opening new establishments. Oberfield et al. (2024) studies firms' decisions of where to locate establishments across heterogeneous locations, but abstracts from markup heterogeneity. Becker et al. (2024) allows for markup heterogeneity across locations but does not study firm size-dependent policy or match our finding that each successive establishment at a firm tends to be smaller relative to its market. We also contribute to this literature by demonstrating empirically what is often hypothesized to be the case in models of local competition: size-dependent markups are set based on the size of a firm's establishments rather than the firm's number of establishments.

Size-dependent markups and misallocation. Our theoretical results build on previous work on the inefficient distortions associated with size-dependent markups in a market equilibrium. Dixit and Stiglitz (1977) study a model with single-establishment identical firms, non-CES demand (with CES as a special case), and a free entry condition. Zhelobodko et al. (2012) and Dhingra and Morrow (2019) include productivity heterogeneity and selection through a fixed cost. Behrens et al. (2020) add multiple sectors, where aggregation across sectors is CES for most of the analysis. These papers all argue that non-CES aggregation—which maps into size-dependent markups—distorts the intensive margin (production at each firm) and the extensive margin (number of firms). We differ in that we add a *firm-specific* entry choice (number of establishments) subject to a nonlinear cost, where aggregation across all establishments both within and across firms is non-CES. If we had a linear cost for opening establishments, for example, then all firms' marginal establishments would be the same size, so there would be no misallocation of establishments across firms. We also simplify by focusing on the case where an establishment's demand elasticity is weakly falling in its output, whereas the referenced papers allow for a more general relationship.

Our results on the costs of these distortions and the policy implications relate to Edmond et al. (2023). They study the cost of size-dependent *firm* markups in a model of single-establishment firms. They show that firm size-dependent policy can eliminate these costs by giving a relative subsidy to larger firms who set higher markups. By contrast, we find that firm size-dependent policy is not effective for undoing the costs of size-dependent *establishment* markups because intensive and extensive margin distortions are not well aligned.

In this context, perhaps the paper most closely related to ours is Afrouzi et al. (2023), who study a similar model but where the extensive margin is a firm's number of customers rather than number of establishments. A major difference is that they suppose all a firm's customers are identical, whereas we match our empirical finding that each successive establishment at a firm is smaller. Moreover, our focus is different in that we study firm size-dependent policy, which they do not. Declining size across a firm's successive establishments is important for our results in this context; otherwise, firm size-dependent policy is much more effective because intensive and extensive margin distortions are better aligned.

The paper proceeds as follows. In Section 2, we describe our empirical analysis. In Section 3, we develop the model and prove theoretical results. In Section 4, we calibrate the model, which we use in Section 5 to study misallocation and firm size-dependent policy.

2 Empirical Analysis

2.1 Data

We use data on the universe of Swedish firms and establishments in services industries from 1997 to 2017. We merge information from two administrative datasets based on firms' tax forms. Both are from Statistics Sweden, the Swedish government agency responsible for producing official statistics. The first data set, Företagens Ekonomi, contains annual financial accounts of all Swedish firms. We use firms' annual sales and expenditures on intermediate inputs. The second data set, Registerbaserad arbetsmarknadsstatistik, contains the complete set of employer-employee linkages at a monthly frequency. For each worker, the data include their labor earnings and the IDs of the establishment and firm at which they are employed. We aggregate across workers within each establishment-year and merge with our firm-level data. This yields annual data on each firm's number of establishments and on each establishment's firm ID, wage bill, employment, municipality, and industry.

There are 291 municipalities in 2017, ranging from Stockholm with about 1 million workers in services industries to Bjurholm with 960 workers in services industries. Industries are at the 5-digit level, which is the finest level of disaggregation available. There are 423 services industries, which have an average of 422 firms each.[5] To restrict attention to services industries, we only include firms and establishments in these industries, respectively, for our firm- and establishment-level analyses. We also only include firm-year observations with at least one employee and positive sales operating in the private economy. This leaves about 3.5 million firm-year observations, which cover about 60% of sales in the Swedish economy. Finally, we deflate nominal variables to 2017 SEK using the GDP deflator.

2.2 Intensive vs. Extensive Margins of Sales

We first show the importance of sales per establishment (the intensive margin) and number of establishments (the extensive margin) as determinants of services firms' sales. Table 1 compares single- and multi-establishment firms. The latter are only 2.7% of services firms, but earn about half the sales in services industries. Multi-establishment firms are so much larger than single-establishment firms because, on average, they sell almost five times as much at each establishment and have about seven times as many establishments.

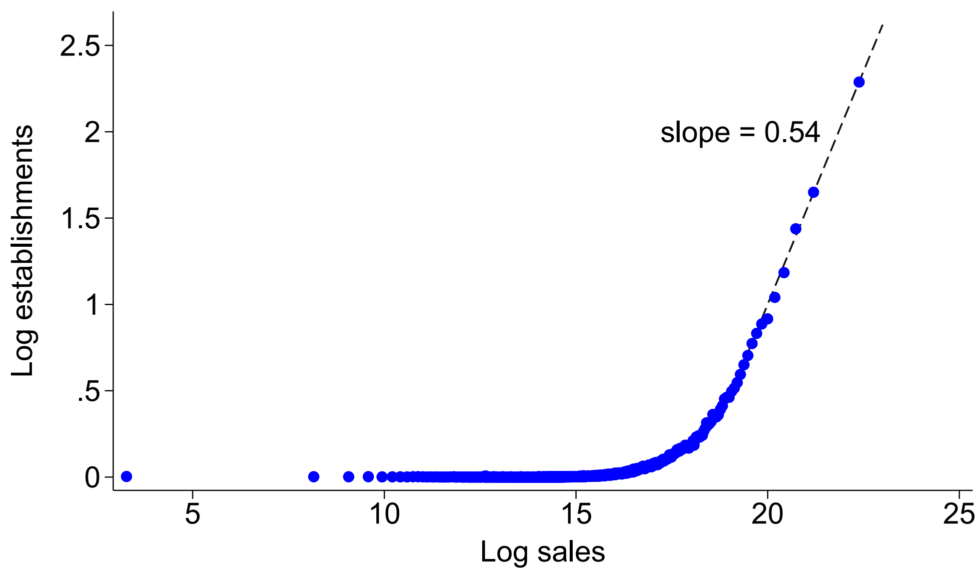
[5] Out of 821 industries, we keep those coded 45 and above, excluding "financial and insurance activities".

Table 1: Single vs. multi-establishment firms

Multi-establishment firms		**Decomposing multi-establishment firm sales**	
Firm share	*Sales share*	*Relative sales per establishment*	*Relative number of establishments per firm*
2.7%	48.7%	4.7	7.2

The first two entries are the share of services firms that are multi-establishment, and the share of services sales earned by multi-establishment firms. The third entry is mean sales per establishment (deflated to 2017 SEK) across multi-establishment firms' establishments relative to across single-establishment firms' establishments. The fourth entry is mean establishments per firm across multi-establishment firms. We compute these in each year, and then take the average across years.

Figure 1: Drivers of firm sales



We group all firm-year observations into 1000 equally sized bins according to log sales (in 2017 SEK). Each dot plots average log number of establishments within a bin against average log sales.

Figure 1 shows the role of the intensive and extensive margins across the firm size distribution. It plots the average log number of establishments at a firm as a function of its log sales. For small firms, the line is flat at 0 because nearly all small firms are single-establishment. For larger firms, the slope of the line is positive, increasing, and converges to about 0.54. This means larger firms tend to have more establishments and tend to sell more at each establishment. Among the largest firms, 54% of variation in sales comes from variation in their number of establishments, and the remaining 46% comes from variation in their sales per establishment.

2.3 Declining Size at Successive Establishments

To study firms' extensive margin decisions (how many establishments to open) we want to know about their *marginal* establishments—those they will most readily open/close—rather than their *average* establishments. To this end, we compare firms' successive establishments, i.e., we ask how the size of a firm's n^{th} establishment depends on n, which we call the establishment's "order". We use employment (number of employees) and the wage bill to measure establishment size because these are the measures for which we have establishment-level data.

We find that each successive establishment at a firm is smaller, conditional on establishment age, and closes more quickly. We interpret this as evidence that as a firm expands, it works its way down to successively worse establishment opportunities. First, the left panel of Figure 2 plots the average log wage bill across an establishment's life cycle conditional on survival for a firm's first five establishments. We restrict attention to firms with at least five establishments, so the set of firms is the same for each line. For each successive establishment, the entire life cycle profile of the wage bill and employment is shifted down. For example, at an *establishment* age of ten years, the wage bill at a firm's third establishment tends to be about 10.5% smaller than the wage bill was at the firm's second establishment. The right panel of Figure **??** plots the probability of survival across establishment age for a firm's first five establishments, again restricting attention to firms with at least five establishments so that the set of firms is the same for each line. For each successive establishment, the entire age profile of survival probabilities is shifted down. In Appendix A, we show that the same stark patterns hold for employment and for firms with different numbers of establishments.

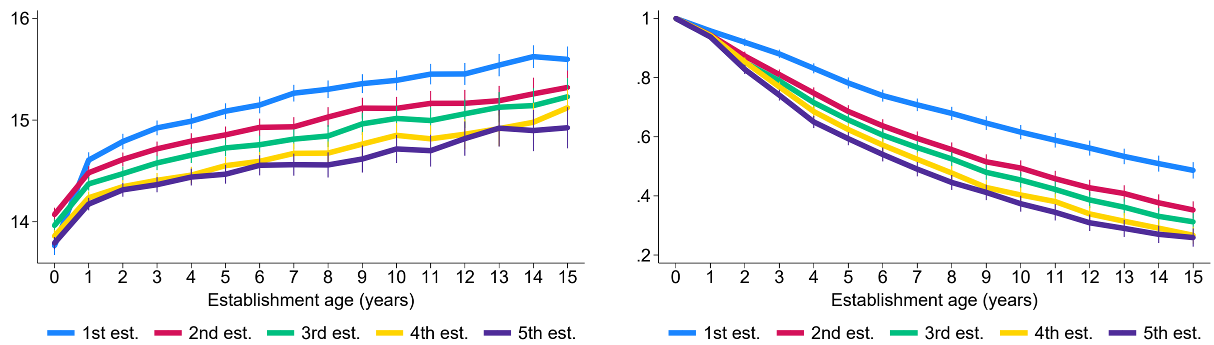
We now conduct a more general parametric analysis, and confirm our findings, using the following regression:

$$\ln(\text{establishment size}_{i,j,n,t}) = \beta \ln(n) + \text{fixed effects}_{i,j,n,t} + \epsilon_{i,j,n,t}. \tag{1}$$

There is one observation for each establishment-year pair. On the left-hand side is the log

Figure 2: Successive establishments

(a) Log wage bill (b) Survival probability



Left panel: average log wage bill (in 2017 SEK) for firms' first five establishments as functions of establishment age, conditional on survival. Right panel: share of establishments that survive to each age for a firms' first five establishments. Averages are computed across firms with at least five establishments. The vertical bars are 95% confidence intervals.

size in year t of the n^{th} establishment opened by firm i in industry j, measured by the establishment's employment or wage bill. On the right-hand side, our main coefficient of interest is β on the log of the establishment's order, n. In our baseline specification, we include establishment age, firm, and year fixed effects. The residual is $\epsilon_{i,j,n,t}$. Before, we restricted attention to firms with at least five establishments to eliminate selection effects. In our regression, firm fixed effects make this unnecessary. As such, we include all establishments.

Columns 1 and 3 in Table 2 show the estimated coefficient β from our baseline regression, using employment and the wage bill, respectively, to measure establishment size. For both size measures, we estimate $\beta \approx -0.2$, so a 1% increase in establishment order is associated with a 0.2% decrease in establishment employment and the wage bill. As an example, this implies that a firm's third establishment is about 7.8% smaller than its second establishment, where the size of each is relative to what we expect given establishment age and year.[6] For both size measures, the estimated β is highly statistically significant.

Columns 2 and 5 in Table 2 add fixed effects for an establishment's municipality (location) crossed with its 5-digit industry. The results are nearly identical to the results from our baseline regression. This suggests that the decline in size of firms' successive establishments holds both in absolute terms as well as relative to the average size in each establishment's

[6] Going from the 2^{nd} to the 3^{rd} establishment, log establishment size falls by $0.2(\ln(3) - \ln(2)) \approx 0.081$.

Table 2: Size of successive establishments

	Log employment			*Log wage bill*		
β	-0.201	-0.197	-0.196	-0.198	-0.195	-0.194
	(0.021)	(0.020)	(0.020)	(0.020)	(0.019)	(0.020)
Fixed effects						
Establishment age	✓	✓	✓	✓	✓	✓
Firm	✓	✓	✓	✓	✓	✓
Year	✓	✓	✓	✓	✓	✓
Municipality-industry		✓	✓		✓	✓
Firm age			✓			✓
R^2	0.840	0.861	0.861	0.740	0.756	0.756
Observations	6,925,089	6,914,070	6,914,070	6,925,089	6,914,070	6,914,070

Estimates of the coefficient, β, on log establishment order from regression equation (1). Columns 1-3 use employment (number of employees) to measure establishment size and columns 4-6 use the wage bill (in 2017 SEK). Check marks indicate which fixed effects each regression includes. We cluster standard errors (in parentheses) at the firm level.

municipality-industry. In this sense, each successive establishment at a firm is smaller relative to its market. Finally, columns 3 and 6 add firm age fixed effects and again show the same results. As such, declining size across successive establishments is not due to firms opening later establishments when they are older.

Finally, we use these results to estimate the relationship between a firm's number of establishments and the size of its marginal establishment relative to its municipality-industry. Specifically, there are two drivers of a firm's marginal establishment relative size. First, the estimated firm fixed effect shifts the size of all a firm's establishments. Second, given a firm fixed effect, firms with more establishments have smaller implied marginal establishments because each successive establishment is smaller. Thus, we regress the estimated firm fixed effect on a firm's log number of establishments. A positive coefficient indicates that firms with more establishments tend to have larger establishments, conditional on the establishment's order, i.e., fixing n, firms with more establishments tend to have a larger n^{th} establishment. A coefficient less than our estimated β—the rate at which establishment size declines with establishment order—indicates that firms with more establishments tend to have smaller marginal establishments because their fixed effect is not enough to compensate for their number of establishments.

Table 3: Marginal establishment size

	Log employment			*Log wage bill*		
	0.202	0.170	0.165	0.167	0.180	0.173
	(0.014)	(0.015)	(0.015)	(0.015)	(0.016)	(0.017)
Fixed effects						
Year	✓	✓	✓	✓	✓	✓
Municipality-industry		✓	✓		✓	✓
Firm age			✓			✓
R^2	0.041	0.664	0.666	0.028	0.711	0.713
Observations	59,114	56,808	56,808	59,114	56,808	56,808

Estimates of the coefficient on log establishment order from regressing the firm fixed effects from Table 2 on log establishment order and the relevant fixed effects. See Table 2 and the text for details. Columns 1-3 use employment (number of employees) to measure establishment size and columns 4-6 use the wage bill (in 2017 SEK). Check marks indicate which fixed effects each regression includes. We cluster standard errors (in parentheses) at the firm level.

The results are in Table 3. We run a separate regression for each regression in Table 2, which produces a different set of firm fixed effect estimates. In each case, we restrict attention to multi-establishment firms and use the same fixed effects as in the original regression, leaving out establishment age, which is not measured at the firm-level, and using a firm's modal municipality-industry. We exclude single-establishment firms because we want to use these results to infer, for our model, the relationship between a firm's number of establishments and its marginal cost of opening an establishment. Thus, we only want firms whose extensive margin is active in the sense that their marginal benefit of opening an establishment is equal to the marginal cost.

We consistently find estimates below our estimated β, except when we use employment to measure establishment size and do not control for municipality-industry or firm age fixed effects. However, the difference between the two is not statistically significant at a high level.

2.4 Size-Dependent Markups

Previous work finds that *within industries* larger firms set higher markups.[7] We now investigate this relationship in services industries in Sweden. In particular, we estimate the relationship between a firm's markup and the two margins of its sales: its sales per establishment and its number of establishments.

We first discuss our estimation strategy. To proxy for a firm's relative markup, we use the ratio of its nominal sales to its expenditures on intermediate inputs. The idea is that if a firm chooses intermediate inputs flexibly and takes the intermediate input price as given, then their sales-to-intermediates ratio is equal to their markup divided by the elasticity of their output with respect to intermediates. We then assume that this output elasticity is the same for all firms in each industry-year. Thus, within an industry-year, the difference between two firms' log markups is equal to the difference between their log sales-to-intermediates ratios.[8]

Using our proxy for firms' relative markups, we run the following regression:

$$\ln(\underbrace{\text{sales/intermediates}_{i,j,t}}_{\text{relative markup}}) = \beta \ln(\text{size}_{i,j,t}) + \text{fixed effects}_{i,j,t} + \epsilon_{i,j,t}. \tag{2}$$

The unit of observation is a firm-year pair. On the left-hand side is our proxy for the relative markup of firm i in industry j and year t: its sales-to-intermediate ratio. On the right-hand side, the coefficient of interest, β, is on the log of some measure of firm size, which can be any subset of total sales, sales per establishment, and number of establishments. Based on our above discussion, we include industry-year fixed effects so that a firm's sales-to-intermediates ratio and size are relative to industry-year averages. Then, β is the elasticity of a firm's *relative* markup with respect to its *relative* size. Finally, $\epsilon_{i,j,t}$ is the residual.

Table 4 reports the estimation results. For our baseline regressions, we restrict attention to firm-year observations with more than one establishment so that when we use a firm's number of establishments to measure its size, the estimated coefficient β does not capture differences between single- and multi-establishment firms. In column 1, we measure a firm's size by its total sales. We find that within an industry-year, firms with higher sales tend to set higher markups. The point estimate, which is highly statistically significant, says that a firm with 1% higher sales has about a 0.06% higher markup. As an example, if firm A's

[7]For example, Edmond et al. (2023) studies size-dependent markups in US manufacturing data, Burstein et al. (2024) use French data, and Afrouzi et al. (2023) use Nielsen scanner data.

[8]For thorough discussions of this approach to estimating relative markups, see De Loecker and Warzynski (2012) and De Ridder et al. (2025).

Table 4: Size-dependent markups

	Baseline regressions				Robustness checks	
	Log markup				*Log markup*	
Log sales	0.062					
	(0.006)					
Log sales per estab.		0.087		0.087	0.186	0.047
		(0.009)		(0.009)	(0.002)	(0.008)
Log number of estabs.			0.021	0.008	-0.107	0.010
			(0.007)	(0.007)	(0.004)	(0.007)
Only multi-establishment	✓	✓	✓	✓		✓
Industry-year fixed effects	✓	✓	✓	✓	✓	✓
IV for sales per estab.						✓
R^2	0.542	0.545	0.532	0.545	0.376	0.547
Observations	93,164	93,164	93,164	93,164	3,454,250	86,272

Each column reports the results from estimating regression equation (2) using different measures of firm size. In each case, we report the estimated coefficients for the included firm size measures. Check marks indicate whether we only include multi-establishment firm-year observations or use all observations, whether we include industry-year fixed effects, and whether we instrument for a firm's sales per establishment using its previous year's value (see Appendix A for details). We cluster standard errors (in parentheses) at the firm level.

markup is 1.2 and industry competitor firm B has twice the sales of firm A, then this implies that firm B's markup is about 1.25. By comparison, using US manufacturing data, Edmond et al. (2023) estimate a coefficient of 0.031, so half as steep a relationship between a firm's relative sales and relative markup.[9]

Our main result on size-dependent markups is in columns 2-4 in Table 4. Together, they demonstrate that this positive relationship between a firm's size and its markup is driven by firms' sales per establishment, not their number of establishments. Column 2 uses a firm's sales per establishment as the size measure in regression equation (2), column 3 uses a firm's number of establishments, and column 4 uses both.[10] When we consider each size measure separately, a firm's markup has a positive statistically significant relationship with the firm's

[9]They report this result in Table C4 of the online appendix.

[10]For column 4, $\beta \ln(\text{size}_{i,j,t})$ on the right-hand side of regression equation (2) becomes $\beta_1 \ln(\text{sales per establishment}_{i,j,t}) + \beta_2 \ln(\text{number of establishments}_{i,j,t})$.

sales per establishment (column 2) and with its number of establishments (column 3). But a 1% increase in sales via sales per establishment implies about four times as big an increase in the markup as a 1% increase in sales via number of establishments. Moreover, when we include both size measures, the coefficient on sales per establishment is unaffected and still highly statistically significant, whereas the coefficient on number of establishments falls by more than half and is no longer statistically significant at the 90% level. Finally, the R^2 is higher when we use sales per establishment to measure size than when we use total sales or number of establishments, and including number of establishments along with sales per establishment does not increase the R^2.

Columns 5 and 6 in Table 4 report the results of two robustness checks that confirm our main finding. In both cases, we use both sales per establishment and number of establishments to measure firm size, so they are comparable to column 4. Column 5 uses all firm-year observations rather than just those with more than one establishment. This increases the coefficient on sales per establishment, which suggests that the relationship between a firm's sales per establishment and its markup is steeper among single-establishment firms than among multi-establishment firms. As a consequence, the regression model predicts too high markups for multi-establishment firms, so the coefficient on number of establishments becomes negative to compensate. Next, column 6 instruments for a firm's sales per establishment using its previous year's value (the first stage results are in Appendix A). This accounts for the possibility that noise in sales mechanically raises the markup and sales per establishment, leading to a positive estimated relationship. The coefficient on sales per establishment falls but remains positive and highly statistically significant. This suggests that there is some noise in sales driving the high coefficient on sales per establishment in our baseline regression, but this noise does not fully account for the positive relationship between a firm's sales per establishment and its markup.

3 Model and Qualitative Results

We now develop the model, discuss its solution, and characterize inefficient distortions in the market equilibrium.

3.1 Model

There is a single period. A representative household consumes the numeraire final good, inelastically supplies labor (the only input) in a perfectly competitive market, and owns all firms. There is a unit mass of firms. Each firm has an initial unit measure of establishments and chooses which additional establishments to open as well as how much to produce at each establishment. Each establishment has a different quality and opening cost, which are known ahead of time, and each firm faces a different distribution of qualities and opening costs. Through each of its establishments, a firm uses labor to produce a differentiated variety, where the establishment's quality serves as a demand shifter. Perfectly competitive final good producers aggregate all the establishment varieties into the final good, which they sell to the household. We leave functional forms relatively general for our theoretical results. We then impose more structure in our calibrated model.

Representative household. The representative household maximizes final good consumption C subject to its budget constraint:

$$C \leq W\bar{L} + \Pi,$$

where W is the wage, $\bar{L}$ is the inelastic labor supply, Π are profits from firms, and the final good price is normalized to 1. We take final good consumption C as our measure of welfare.

Final good producers and demand. Perfectly competitive final good producers aggregate varieties from firms' establishments into final good output Y. They sell the final good to the representative household, so their output must equal final good consumption:

$$C = Y.$$

Specifically, final good producers purchase varieties from a double continuum of establishments: each firm $i \in [0, 1]$ sells a different variety at each of its establishments $n \in [0, N(i)]$. If final good producers purchase $y(i, n)$ from firm i's establishment n, then final good output Y is given implicitly by the Kimball (1995) aggregator:

$$\int_0^1 \int_0^{N(i)} \varphi(i,n)\Upsilon(q(i,n))dndi = 1 \qquad q(i,n) \equiv \frac{y(i,n)}{Y}, \tag{3}$$

where $\varphi(i,n)$ is the quality of firm i's establishment n, $q(i,n)$ is the relative real output of firm i's establishment n, and $\Upsilon(\cdot)$ is twice continuously differentiable, strictly increasing, and strictly concave. That is, given variety purchases $y(\cdot,\cdot)$, final good output Y is such that (3) holds. The aggregation technology is constant returns-to-scale because multiplying Y and each $y(i,n)$ by the same positive constant leaves (3) unchanged. A special case of the aggregator is CES (constant elasticity of substitution), where $\Upsilon(q) = q^{\frac{\bar{\sigma}-1}{\bar{\sigma}}}$ for some $\bar{\sigma} > 1$.

Final good producers choose demand $y(\cdot,\cdot)$ and final good output Y to maximize profits

$$Y - \int_0^1 \int_0^{N(i)} p(i,n) y(i,n) dn di$$

subject to the aggregation technology (3), where $p(i,n)$ is the price at firm i's establishment n. The resulting relative demand for each variety, $q(i,n) \equiv y(i,n)/Y$, is given by[11]

$$p(i,n) = \varphi(i,n)\Upsilon'(q(i,n))D \qquad D \equiv \left(\int_0^1 \int_0^{N(i)} \varphi(i,n) q(i,n) \Upsilon'(q(i,n)) dn di \right)^{-1}, \tag{4}$$

where $D > 0$ is a demand index. Given an establishment's relative output $q(i,n)$, its price $p(i,n)$ moves one-for-one with its quality $\varphi(i,n)$. Given quality, an establishment's price is falling in its relative output because $\Upsilon(\cdot)$ is strictly concave. Finally, an establishment's price does not depend on output at its firm's other establishments because it only depends on other establishments through *economy-wide* aggregates, D and Y.

Firms, establishments, and production. Each firm, indexed by $i \in [0,1]$, begins with a unit measure of establishments, indexed by $n \in [0,1]$, which we interpret as their first establishment in our continuous setting. They then face an infinite set of additional establishment opportunities $n \in [1,\infty)$. The quality at firm i's establishment n is $\varphi(i,n)$, which is defined for all $n \geq 0$. The cost of opening firm i's establishment n is $\kappa(i,n)$ units of labor, which is defined only for $n \geq 1$. These functions are common knowledge, so firm i faces no risk when choosing which additional establishments to open. Productivity is the same at all establishments: if firm i opens establishment n, then output at that establishment is the

[11]The first order condition for $y(i,n)$ is $p(i,n) \geq (\lambda/Y)\varphi(i,n)\Upsilon'(y(i,n)/Y)$, where λ is the Lagrange multiplier on the aggregation constraint and where the inequality holds with equality if $y(i,n) > 0$. For $y(i,n) = 0$, the inequality can be strict if $\Upsilon'(0) < \infty$ (unlike with CES) because then there is a finite price threshold above which demand is 0. Without loss of generality, we suppose the price is weakly below this threshold. Multiply both sides of the first order condition by $y(i,n)$, integrate across establishments, and plug in that final good producer profits are zero (implied by perfect competition) to get $Y = \lambda/D$. Plug back into the first order condition to get (4).

labor used for production, $l(i,n)$. Hence, the marginal cost of production is the wage W.

Without loss of generality and for ease of exposition, we suppose establishments are ordered so that in equilibrium, firm i's value of opening establishment $n \geq 1$ is decreasing in n.[12] For ease of exposition, but with some loss of generality, we also suppose that for each firm i, $\varphi(i,n)$ is continuous for all $n \geq 0$ and $\kappa(i,n)$ is continuous for all $n \geq 1$.

Our ordering of establishments means that if firm i opens a measure $N(i)$ of establishments, then it opens establishments $n \in [0, N(i)]$, which we already implicitly took as given in the final good producer problem above. Thus, firm i's establishment opening decision is just a decision of what measure of establishments to open, $N(i) \geq 1$. If firm i does not open additional establishments—sets $N(i) = 1$—then we say it is a "single-establishment" firm. Otherwise, if firm i sets $N(i) > 1$, then we say it is a "multi-establishment" firm.

Each firm i chooses its measure $N(i)$ of establishments as well as the price $p(i,n)$, labor $l(i,n)$, and output $y(i,n)$ for each establishment $n \in [0, N(i)]$ to maximize profits:

$$\int_0^{N(i)} [p(i,n)y(i,n) - Wl(i,n)]dn - W \int_1^{N(i)} \kappa(i,n)dn$$

subject to the production function, $y(i,n) = l(i,n)$, the demand curve (4), and the establishment quality function $\varphi(i,\cdot)$. The first integral in firm i's profits is the production profits from its measure $N(i)$ establishments. The second integral is the cost of opening establishments beyond the initial unit measure.

Establishment size and markups. To solve a firm's profit maximization problem, we first characterize its optimal production and pricing decisions at each establishment. Decisions are independent across a firm's establishments because a firm is small relative to the economy and establishments are only related through economy-wide aggregates. Hence, if firm i's establishment n has strictly positive sales, then its markup of price over marginal cost, $\mu(i,n) \equiv p(i,n)/W$, must satisfy the usual expression:

$$\mu(i,n) = \frac{\sigma(q(i,n))}{\sigma(q(i,n)) - 1} \qquad \sigma(q(i,n)) \equiv \frac{-\Upsilon'(q(i,n))}{q(i,n)\Upsilon''(q(i,n))} = \frac{-p(i,n)}{y(i,n)} \frac{\partial y(i,n)}{\partial p(i,n)}, \tag{5}$$

where $\sigma(q(i,n))$ is the demand elasticity (of quantity with respect to price) at firm i's establishment n, which depends on the establishment's relative output $q(i,n) \equiv y(i,n)/Y$.

[12] In equilibrium, the value of opening an establishment is the optimal profits from that establishment minus the opening cost, where an establishment's optimal profits do not depend on the firm's choice of which other establishments to open.

To finish solving the optimization problem at each establishment, we make the following assumption that the demand elasticity is sufficiently high and is weakly decreasing in an establishment's size.

Assumption 1. *For all q, $\sigma(q) \equiv \frac{-\Upsilon'(q)}{q\Upsilon''(q)}$ is weakly decreasing. Moreover, there exists a $\bar{q} > 0$ such that $\sigma(q) > 1$ for all $q \leq \bar{q}$.*

Assumption 1 implies that given an establishment's quality $\varphi(i, n)$, there is a unique relative output and price pair that satisfies expression (5) and demand curve (4).[13] Furthermore, an establishment's sales, price, markup, and profits are weakly increasing in its quality. The restriction that larger establishments face a weakly lower demand elasticity—and so set weakly higher markups—matches our empirical results on size-dependent markups.

Measure of establishments. We next characterize each firm's optimal measure of establishments. Let $\pi(\varphi)$ be the profits of an establishment with quality φ. Then, firm i's benefit of opening its marginal establishment (increasing its measure $N(i)$ of establishments) is the profits it will earn from that establishment, $\pi(\varphi(i, N(i)))$. The cost of opening the marginal establishment is $W\kappa(i, N(i))$. Our ordering of firm i's establishments then means that for $n \geq 1$, $\pi(\varphi(i, n))/\kappa(i, n)$ is decreasing in n. We now make the following assumption to ensure that firm i has a unique and finite optimal measure of establishments.

Assumption 2. *For each firm i, $\pi(\varphi(i, n))/\kappa(i, n)$ is strictly decreasing in n for $n \geq 1$, and goes to 0 as n goes to infinity.*

Assumption 2 implies that firm i does not open any establishments beyond its initial unit measure—it is a "single-establishment" firm—if the profits at its establishment $n = 1$ are sufficiently low relative to the opening cost, i.e., if $\pi(\varphi(i, 1)) \leq W\kappa(i, 1)$. On the other hand, if $\pi(\varphi(i, 1)) > W\kappa(i, 1)$, then firm i is a "multi-establishment" firm and chooses the unique $N(i) > 1$ that satisfies

$$\pi(\varphi(i, N(i))) = W\kappa(i, N(i)). \tag{6}$$

[13] The second part of Assumption 1 is necessary because expression (5) implies that all establishments with strictly positive relative output q set q such that they face a demand elasticity $\sigma(q)$ strictly above 1. As such, if the second statement is violated, then the aggregator constraint (3) cannot be satisfied. We suppose $\bar{q}$ is sufficiently high so that there is an equilibrium.

Firm size and markups. To compute firm-level outcomes, we aggregate across a firm's establishments. Firm i's sales are $S(i) = \int_0^{N(i)} p(i,n)y(i,n)dn$ and its production labor is $L(i) = \int_0^{N(i)} l(i,n)dn$. We define a firm's markup as its sales over production costs in order to align with our definition of an establishment's markup, which is the same ratio. Hence, a firm's markup is the cost-weighted average of its establishment markups:

$$\mu(i) \equiv \frac{S(i)}{WL(i)} = \int_0^{N(i)} \frac{l(i,n)}{L(i)} \mu(i,n)dn.$$

Aggregation. The economy aggregates so that final good output is

$$Y = ZL_p,$$

where $L_p = \int_0^1 L(i)di$ is labor used for production (rather than for opening establishments) and Z is aggregate productivity:[14]

$$Z = \left(\int_0^1 \int_0^{N(i)} q(i,n)dndi \right)^{-1}. \tag{7}$$

Hence, aggregate productivity depends on each firm's measure of establishments and its relative output at each establishment. The distribution of establishment qualities then affects aggregate productivity through relative outputs. For example, if all establishment qualities double and establishment purchases $y(\cdot,\cdot)$ remain constant, then final good output Y must rise until the Kimball aggregation constraint (3) is again satisfied. As a result, relative output $q(i,n)$ is lower at all establishments, so aggregate productivity Z is higher.

Equilibrium. In equilibrium, the representative household maximizes consumption C subject to its budget constraint, taking as given wage income $W\bar{L}$ and firm profits Π. Each firm i chooses their measure of establishments $N(i)$ and the price $p(i,n)$, labor $l(i,n)$, and output $y(i,n)$ at each establishment to solve their profit maximization problem, taking as given final good output Y, the wage W, and the demand index D. Final good producers choose demand $y(i,n)$ for each establishment, and final good output Y to solve their profit maximization problem, taking as given the price $p(i,n)$ at each establishment. Perfect competition implies

[14]Write $Z^{-1} = L_p/Y = \int_0^1 \int_0^{N(i)} (l(i,n)/Y)dndi$. Using $y(i,n) = l(i,n)$ and $q(i,n) = y(i,n)/Y$ yields (7).

that final good producer profits are zero.

Household consumption must equal final good output: $C = Y$. Final good output Y and demand for each variety $y(\cdot,\cdot)$ must satisfy aggregation (3). Demand for each variety must equal output, which must equal the labor used to produce it: $y(i,n) = l(i,n)$. The labor used by firms for production and opening establishments must equal the household's inelastic labor supply:

$$\int_0^1 L(i)di + \int_0^1 \int_1^{N(i)} \kappa(i,n)dndi = \bar{L}.$$

We can find an equilibrium as follows. Given final good output Y and the demand index relative to the wage D/W, use the demand curve (4) and the optimal markup expression (5) to get establishment relative output and profits relative to the wage as functions of establishment quality. The latter implies each firm's measure of establishments $N(i)$ using (6). Then check whether the Kimball aggregator constraint holds and the labor market clears. Hence, there are two aggregate variables (Y and D/W) that must satisfy two equations (Kimball aggregator constraint and labor market clearing). We can then compute the remaining equilibrium variables.

3.2 Inefficient Distortions in the Market Equilibrium

We now provide a general characterization of the inefficient distortions to firms' decisions in a market equilibrium. We later calibrate the model to compute the costs of these distortions and to assess the usefulness of firm size-dependent policy for reducing these costs. The market equilibrium can be inefficient due to misallocation of the fixed labor supply across its various uses—opening establishments at each firm and producing at each establishment. On the margin, we can raise welfare (household consumption) by shifting labor from uses with a low marginal social value to uses with a high marginal social value. Hence, to state our result, we define $V_{prod}(i,n)$ to be the marginal social value of labor used to produce at firm i's establishment n and define $V_{estab}(i)$ to be the marginal social value of labor used to open and produce at firm i's marginal establishment. Formally, suppose a social planner has an infinitesimal quantity Δ of labor beyond the inelastic labor supply $\bar{L}$. Hold fixed the allocation of the $\bar{L}$ labor across its various uses from the market equilibrium. If the planner uses the additional Δ labor to increase production at firm i's establishment n, then household consumption increases by $V_{prod}(i,n)\Delta$. If the planner uses the additional labor to open establishments at firm i and produce the market equilibrium output, $y(i, N(i))$, at those establishments, then household consumption increases by $V_{estab}(i)\Delta$.[15] We can now state our result.

Theorem 1. *If the demand elasticity $\sigma(q)$ is constant in relative output q, then the market equilibrium is efficient.*

If the demand elasticity $\sigma(q)$ is always strictly decreasing in relative output q, then the market equilibrium is inefficient and the following are true:

1. **Misallocation of production.** $V_{prod}(i, n) > V_{prod}(j, m)$ *if firm i's establishment n is strictly larger than firm j's establishment m, i.e., $q(i, n) > q(j, m)$;*

2. **Misallocation of establishments.** $V_{estab}(i) > V_{estab}(j)$ *if firm i's marginal establishment is strictly larger than firm j's marginal establishment, i.e., $q(i, N(i)) > q(j, N(j))$, and firm i is a multi-establishment firm;*

3. **Production vs. establishment opening.** $V_{prod}(i, n) > V_{estab}(j)$ *if firm i's establishment n is weakly larger than firm j's marginal establishment, i.e., $q(i, n) \geq q(j, N(j))$.*

Proof. See Appendix B.1. ■

The theorem first states that if the demand elasticity is constant (CES demand), then we cannot improve welfare relative to the market equilibrium by reallocating labor. Second, if the demand elasticity is falling in an establishment's relative output—as in our calibrated model—then on the margin, we can improve household consumption by 1) reallocating production from small establishments to large establishments, 2) reallocating establishments from firms with small marginal establishments to firms with large marginal establishments, and 3) closing marginal establishments in exchange for producing more at other weakly larger establishments. For 2), the firm with the larger marginal establishment must be a multi-establishment firm because a large marginal establishment at a single-establishment firm may have an arbitrarily high opening cost.

To understand the strength of 3), first note that it is not the strongest possible statement: by continuity, $V_{prod}(i, n) > V_{estab}(j)$ for some establishments that are strictly smaller than

[15]This means the planner opens $\frac{\Delta}{\kappa(i,N(i))+y(i,N(i))}$ establishments because each establishment uses $\kappa(i, N(i))$ in establishment-opening labor and $y(i, N(i))$ in production labor.

firm j's marginal establishment, i.e., with $q(i, n) < q(j, N(j))$. Moreover, in our calibrated model, a firm's marginal establishment is its smallest establishment because we match our empirical finding that each successive establishment at a firm is smaller. In that case, 3) implies that we can improve household consumption by closing establishments at a firm in exchange for producing more *at any of the firm's other establishments* or at many smaller firms' establishments. We later illustrate Theorem 1 in our calibrated model in Figure 5.

Intuitively, the allocation of labor in the market equilibrium is determined by equalizing the marginal private value (the effect on a firm's profits) of the various uses of labor. So uses with a high marginal social value are those that are more undervalued by firms. As such, the marginal social value of using labor to produce at a firm's establishment is high if the firm more deeply undervalues the establishment's *marginal* output. On the other hand, the marginal social value of using labor to open a new establishment at a firm is high if the firm more deeply undervalues the establishment's *average* output.

Now, firms undervalue output at an establishment—relative to the social value—because they face a finite demand elasticity: to sell an additional unit at an establishment, a firm must cut the price on all the establishment's other units. The lower an establishment's demand elasticity, the more the firm must cut its price, and so the more it undervalues additional output. It follows that if the demand elasticity is constant in an establishment's relative output, then firms undervalue marginal output and average output at all establishments by the same amount. As such, the market equilibrium is efficient.

With a demand elasticity that is falling in an establishment's relative output (as in our calibrated model), firms more deeply undervalue each successive unit of output at an establishment. One consequence is that the larger an establishment, the more firms undervalue its marginal output and its average output. Thus, we can improve welfare by reallocating production from small establishments to large establishments and by reallocating establishments from firms with small marginal establishments to firms with large marginal establishments. Another consequence is that firms undervalue an establishment's marginal output more than they undervalue its average output. Thus, we can improve welfare by closing establishments in exchange for producing more at larger (or even somewhat smaller) establishments.

4 Quantifying the Model

We now calibrate the model to match our data on services industries in the Swedish economy from 1997-2017. We then describe key features of the equilibrium and test the model using untargeted data moments. In the following sections, we use the calibrated model to study misallocation and policy.

4.1 Calibration

Establishment opening cost and quality functions. The quality at firm i's first unit measure of establishments, $n \in [0,1]$, is constant at $\varphi(i,n) = \bar{\varphi}(i)$, which we call firm i's baseline quality. Then, quality at firm i's additional establishments, $n \geq 1$, declines with elasticity $\rho > 0$, so it is given by $\varphi(i,n) = \bar{\varphi}(i)n^{-\rho}$. We impose that each successive establishment at a firm is lower quality so that it is smaller, as we found in the data (and match in our calibration below).

All firms face a constant marginal cost of opening establishments, $\bar{\kappa}$. We do so for a few reasons. First, it is a natural base case, which we are able to consider because each successive establishment a firm opens is lower quality; if all a firm's establishments were the same quality, then we would need firms to face an increasing marginal cost of opening establishments so that their optimization problem would have a solution. Second, this fits our empirical results (Tables 2 and 3), in which we cannot reject that a constant marginal establishment size across firms with different numbers of establishments. Finally, this case makes the results easier to follow, so it is useful for illustrating the key points.

Given these functions, all heterogeneity across firms comes from heterogeneity in their baseline quality, $\bar{\varphi}(i)$, which we suppose is drawn from a Pareto distribution with minimum 1 and tail parameter ω, i.e., the pdf at $\bar{\varphi}$ is $\omega\bar{\varphi}^{-\omega-1}$. Finally, our establishment ordering and Assumption 2 are satisfied by $\rho > 0$.

Kimball Aggregator Function. For the aggregator function $\Upsilon(\cdot)$, we use the Klenow and Willis (2016) specification of the *derivative*:

$$\Upsilon'(x) = \frac{\bar{\sigma}-1}{\bar{\sigma}}\exp\left(\frac{1-x^{\frac{\epsilon}{\bar{\sigma}}}}{\epsilon}\right),$$

where $\bar{\sigma}$ and ϵ are parameters. It follows that the demand elasticity (of quantity with respect to price) and markup at firm i's establishment n are:

$$\sigma(q(i,n)) = \bar{\sigma}q(i,n)^{\frac{-\epsilon}{\bar{\sigma}}} \qquad \mu(i,n) = \frac{\bar{\sigma}q(i,n)^{\frac{-\epsilon}{\bar{\sigma}}}}{\bar{\sigma}q(i,n)^{\frac{-\epsilon}{\bar{\sigma}}}-1}.$$

Hence, $\bar{\sigma}$ is the demand elasticity at relative output 1 and $|\epsilon/\bar{\sigma}|$ is the super elasticity—the elasticity of the demand elasticity with respect to quantity. If $\epsilon = 0$, then we have CES demand, so the demand elasticity and markup are constant in an establishment's relative output. If $\epsilon > 0$, then the demand elasticity is falling in an establishment's relative output, so larger establishments set higher markups. If $\epsilon < 0$, then larger establishments face a higher demand elasticity and set lower markups. Thus, $\epsilon \geq 0$ satisfies Assumption 1 and matches our empirical finding that firms with higher sales per establishment set higher markups.

Our aggregator function differs from Klenow and Willis (2016) in that we pin down $\Upsilon(\cdot)$ by setting $\Upsilon(0) = 0$, whereas they set $\Upsilon(1) = 1$. This distinction is important because if $\epsilon > 0$, then there is a strictly positive quality cutoff below which an establishment has zero sales in equilibrium. If $\Upsilon(0) \neq 0$, which is generically the case with $\Upsilon(1) = 1$, then these non-producing establishments affect economic outcomes. For example, if $\Upsilon(0) > 0$, then opening non-producing establishments raises aggregate productivity. This is particularly relevant when we turn to policy and can incentivize firms to open many low quality establishments. To avoid this oddity, we set $\Upsilon(0) = 0$. The resulting aggregator function is

$$\Upsilon(q) = \frac{\bar{\sigma} - 1}{\epsilon} \int_0^{q^{\epsilon/\bar{\sigma}}} \tilde{q}^{\frac{\bar{\sigma}}{\epsilon} - 1} \exp\left(\frac{1 - \tilde{q}}{\epsilon}\right) d\tilde{q}.$$

Parameters. We set the inelastic labor supply to $\bar{L} = 0.924$ so that final good output is $Y = 1$. This leaves five parameters to calibrate: 1) $\bar{\kappa}$, the establishment opening cost shifter; 2) ω, the Pareto tail parameter for the baseline quality distribution; 3) $\bar{\sigma}$, the demand elasticity at relative output of 1; 4) $\epsilon/\bar{\sigma}$, the demand super elasticity; and 5) ρ, the elasticity with which quality declines across a firm's successive establishments.

To calibrate these parameters, we exactly match four moments in our Swedish data on services industries, plus an estimate of the average markup. The four moments are 1) the share of firms that are multi-establishment; 2) the share of sales earned by multi-establishment firms; 3) the estimated elasticity of the markup with respect to sales per establishment among multi-establishment firms, controlling for industry-year fixed effects (column 2 of Table 4); and 4) the rate at which the log wage bill or log employment falls across a firm's successive establishments (Table 2).[16] For our last data moment, Sandström

[16]For 3) and 4), we run the same regressions on our model data, including firm fixed effects in the latter case. To create establishment-level data, we split a firm's measure of establishments into as fine a grid as possible given our grid for establishment quality.

Table 5: Calibration Targets and Calibrated Parameter Values

Moment	**Data**	**Model**
fraction of firms with multiple establishments	0.027	0.027
sales share of multi-establishment firms	0.49	0.49
log(markup) on log(sales per establishment)	0.087	0.087
average markup	1.25	1.25
wage bill decline across successive establishments	0.20	0.20

Parameter	**Value**
$\bar{\kappa}$ (establishment opening cost shifter)	0.267
ω (baseline quality tail parameter)	13.6
$\bar{\sigma}$ (demand elasticity at relative output of 1)	4.17
$\epsilon/\bar{\sigma}$ (demand super elasticity)	0.324
ρ (establishment quality elasticity)	0.075

(2020) estimates the economy-wide sales-weighted average markup in Sweden from 1997-2017 (the same time period as our data) to be about 1.15.[17] Using the cost share of intermediate inputs as an estimate of its output elasticity in each 2-digit industry-year pair, we estimate a cost-weighted average markup of 1.35. We use 1.25 as a midpoint; our results are nearly the same if we instead use a markup of 1.15 or 1.35. Table 5 lists the data moments and calibrated parameter values.

All five moments jointly determine all five parameters. Nonetheless, there is an intuitive mapping between particular moments and parameters. The demand parameters $\bar{\sigma}$ and ϵ determine the relationship between the markup and relative output at each establishment, so they map into the elasticity of the markup with respect to sales per establishment and the average markup. The rate at which establishment quality declines across a firm's successive establishments, ρ, matches the rate at which the wage bill declines with each successive establishment. The establishment opening cost shifter $\bar{\kappa}$ and the tail parameter ω of the firm baseline quality distribution map into the fraction of firms with multiple establishments and the sales share of multi-establishment firms. Specifically, a higher $\bar{\kappa}$ means firms open fewer establishments, so there are fewer multi-establishment firms and each multi-establishment firm has lower sales. A higher ω pushes these moments in the same direction, but particularly

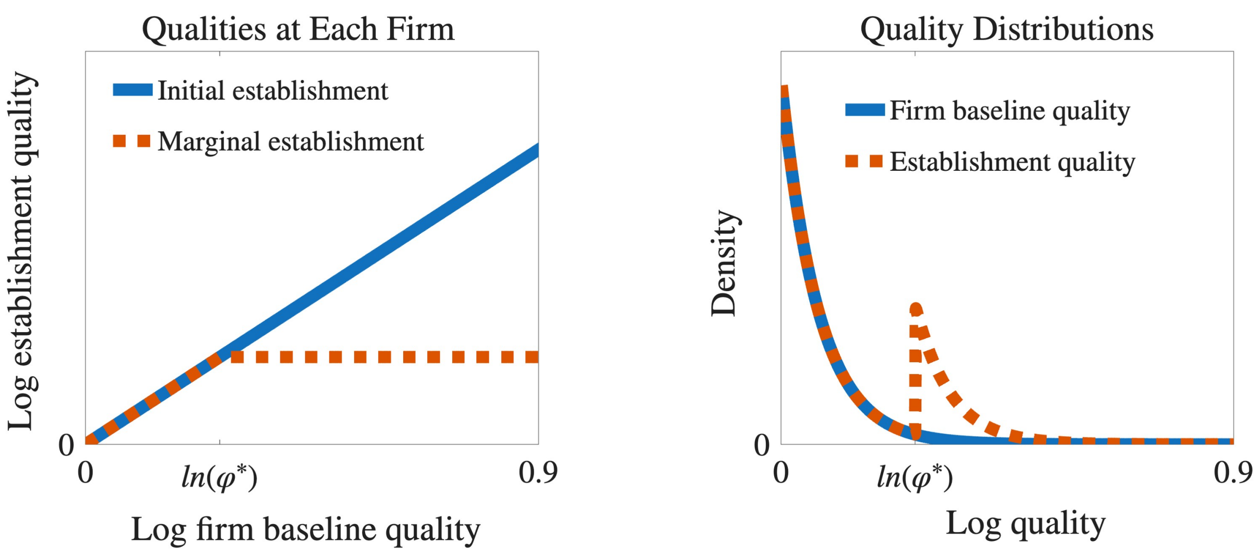
[17] Sandström (2020) does not report the *cost-weighted* average markup, so we use the sales-weighted average. In our calibrated model, both weightings yield the same average markup.

affects the highest quality firms, so it has a stronger effect on the sales share of multi-establishment firms relative to the share of firms with multiple establishments.

4.2 Initial Equilibrium

We now illustrate features of the market equilibrium of our calibrated model. First, all firms face the same constant marginal cost of opening establishments, $W\bar{\kappa}$. It follows from equation (6) for a firm's optimal measure of establishments that there is a firm baseline quality cutoff φ^* given by $\pi(\varphi^*) = W\bar{\kappa}$ such that firm i is a multi-establishment firm if and only if their baseline quality, $\bar{\varphi}(i)$, is above φ^*. Moreover, all firms face the same rate at which their establishment quality declines across their successive establishments ($\rho > 0$). It follows that a firm's optimal measure of establishments is increasing in their baseline quality because the higher a firm's baseline quality, the more establishments they must open until profits equal $W\bar{\kappa}$.

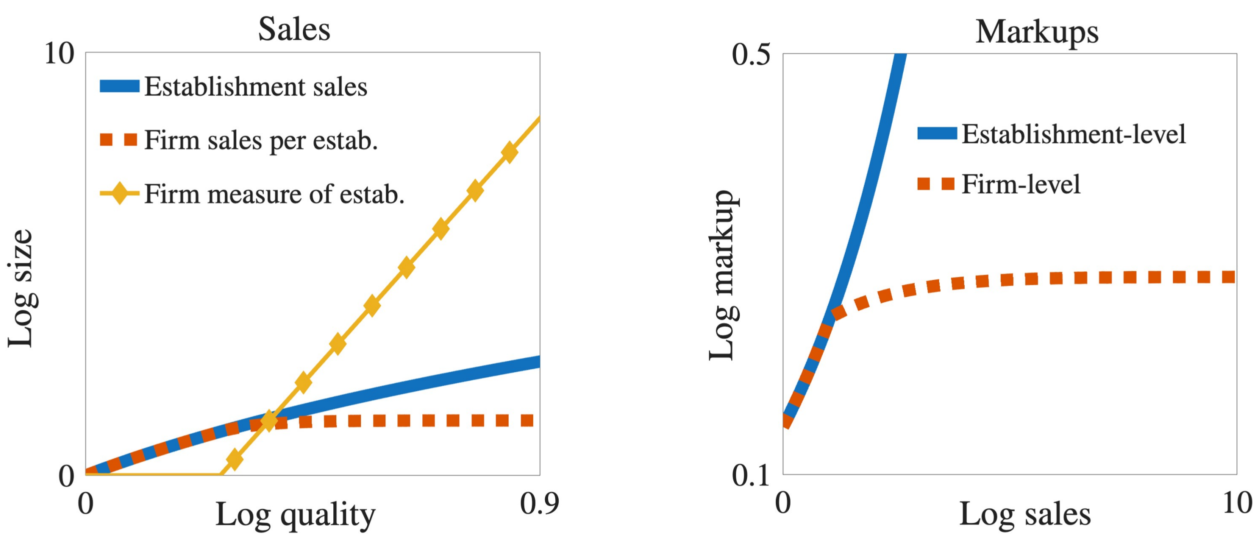
Figure 3: The Set of Establishments



Left panel: log quality at a firm's initial unit measure of establishments and at its marginal establishment, as functions of its log baseline quality. Right panel: the density of the distribution of firm baseline quality and of establishment quality; the former integrates to the measure of firms, which is 1, and the latter integrates to the measure of establishments, which is greater than 1. In both panels, φ^* is the firm baseline quality cutoff above which firms are multi-establishment.

Figure 3 displays the resulting set of open establishments. The left panel shows the interval of establishment qualities at each firm. A firm operates establishments with quality between the two lines, where its initial unit measure of establishments ($n \in [0, 1]$) are its highest quality establishments, and its marginal establishment ($n = N(i)$) is its lowest quality establishment. Above the cutoff φ^*, firms have multiple establishments so there is a gap between the two lines. We can see that all multi-establishment firms have the same quality marginal establishment because they face the same marginal cost of opening establishments. The right panel depicts the resulting distribution of establishment qualities. The density of firm baseline quality, which is exogenously Pareto distributed with tail parameter ω, is also the density of establishment quality across firms' initial unit measures of establishments. The density of establishment quality is endogenous due to firms' establishment opening decisions. The gap between the two densities shows the set of additional establishments firms open beyond their initial unit measures.

Figure 4: Sales and Markups



Left panel: log establishment sales as a function of log establishment quality and log firm sales per establishment and measure of establishments as functions of log firm baseline quality. Right panel: log establishment markup as a function of log establishment sales and log firm markup as a function of log firm sales. In both panels, log sales are relative to log sales at an establishment with log quality 0.

Finally, Figure 4 illustrates sales and markups across firms and their establishments. The left panel shows that higher quality establishments sell more. Then, higher quality firms sell more per establishment because they have higher quality establishments. They also have more establishments, which brings down the average quality of their establishments. As a result, sales per establishment flattens out among multi-establishment firms (with baseline quality above φ^*). This means that variation in sales among larger firms is increasingly driven by variation in measure of establishments rather than in sales per establishment, as in the data (Section 2.2). The right panel shows that larger establishments set higher markups. The same is true for firms, but the relationship flattens out because their sales are increasingly driven by their measure of establishments, which do not affect markups.

5 Quantitative Results

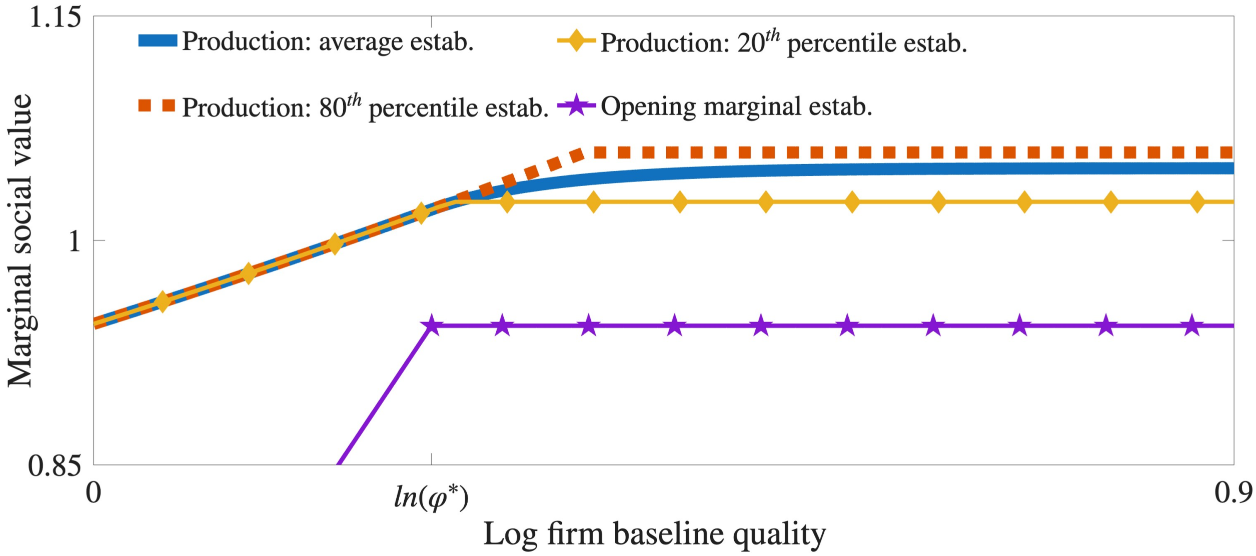
5.1 Misallocation

We now solve a planner problem to compute the costs of inefficient distortions in the market equilibrium of our calibrated model. The planner chooses each firm's measure of establishments and how much to produce at each establishment in order to maximize consumption (equivalently, final good output) subject to the Kimball aggregation technology (3), firms' establishment opening and production technologies, and the inelastic labor supply $\bar{L}$. Therefore, the planner can improve on the market equilibrium by reallocating the fixed labor supply across its various uses.

To build intuition, Figure 5 illustrates the planner's marginal incentives in the market equilibrium, characterized qualitatively in Theorem 1. First, the top three lines depicts $V_{prod}(i, n)$—the marginal social value of production at firm i's establishment n—at each firm's 80^{th} percentile largest establishment, average establishment, and 20^{th} percentile largest establishment. As expected from Theorem 1, the first line sits above the second, which sits above the third, which means the planner wants to reallocate production within firms from smaller establishments to larger establishments. Moreover, each line is weakly increasing because larger firms have larger establishments. As such, the planner wants to reallocate production from small firms' establishments to large firms' establishments. The dispersion in marginal social values is bigger across firms (moving along the x-axis) than within firms (moving along the y-axis), which indicates that the marginal gains from reallocating production are bigger across firms than within firms.

Next, the bottom line of Figure 5 depicts $V_{estab}(i)$—the marginal social value of opening and producing the market equilibrium quantity at firm i's marginal establishment—at each firm. As expected from Theorem 1, this line is constant among multi-establishment firms (with baseline quality above φ^*) because all multi-establishment firms have the same quality marginal establishment. As such, the planner does not want to reallocate establishments

Figure 5: Marginal Social Values in the market equilibrium



The top three lines are $V_{prod}(i,n)$ at firm i's 80^{th} percentile ($n = 0.8N(i)$), average, and 20^{th} percentile ($n = 0.2N(i)$) establishments as functions of firm i's log baseline quality, where the average weights each establishment by its increase in production following a subsidy to firm i's sales. The bottom line is $V_{estab}(i)$ as a function of firm i's log baseline quality. $V_{prod}(i,n)$ and $V_{estab}(i)$ are normalized by aggregate productivity Z, which is the production weighted average of $V_{prod}(i,n)$ across all establishments. φ^* is the baseline quality cutoff above which firms are multi-establishment.

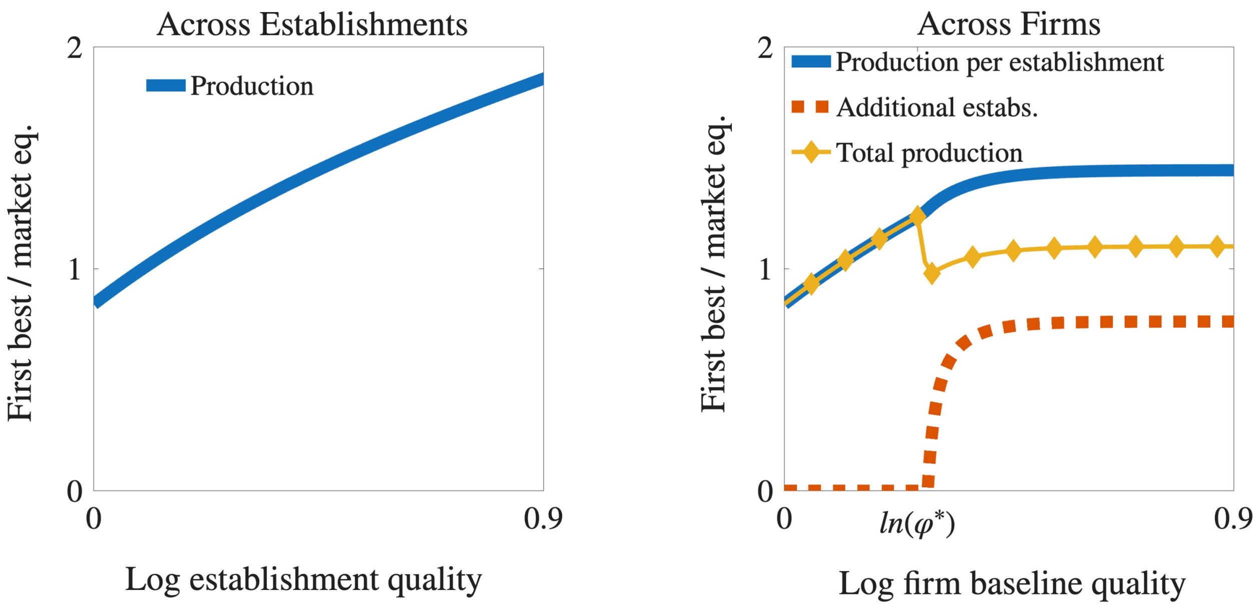
across firms. More significantly, this line sits well below the others because marginal establishments are small and the marginal value of opening an establishment is lower than the marginal value of producing at any larger establishment. Therefore, the planner wants to close establishments in exchange for producing more at essentially any establishment in the economy.

First best allocation. We solve the planner's problem, which yields the first best allocation of the measure of establishments at each firm and production at each establishment. Going from the market equilibrium to the first best increases aggregate consumption by 0.81%. To understand the quantitative significance of this number, it is useful to compare to Edmond et al. (2023), who study a similar model with size-dependent markups but with only single-establishment firms.[18] They find that the static gains from eliminating misallocation are amplified more than three-fold because output is used as an intermediate input and to build

[18]Their version with quality rather than productivity heterogeneity is in online appendix E.1.

capital over time.[19] Thus, we understate the welfare cost of distortions because we exclude these amplifying macroeconomic forces.

Figure 6: Reallocation in the First Best



Left panel: establishment production (labor) in the first best relative to the market equilibrium, as a function of log establishment quality. Right panel: a firm's production (labor) per establishment, measure of additional establishments (beyond firms' initial unit measures), and total production (labor) in the first best relative to the market equilibrium, as functions of its log baseline quality; φ^* is the baseline quality cutoff above which firms are multi-establishment in the market equilibrium.

The 0.81% increase in consumption comes from a 1.8% *fall* in aggregate productivity and a shift of 2.4% of the labor supply from opening establishments to production. Figure 6 illustrates how the planner reallocates labor to generate these gains, which is as expected given marginal incentives in the market equilibrium (see Figure 5 above). The left panel of Figure 6 depicts each establishment's production (equivalently, labor) in the first best relative to the market equilibrium. The line is increasing, which indicates a reallocation of production from small (low quality) establishments to large (high quality) establishments. The line sits mostly above 1 because the planner closes enough establishments that production rises at

[19]Specifically, Edmond et al. (2023) implement a size-dependent tax/subsidy that eliminates markup dispersion, but leaves the aggregate markup unchanged. Initially, aggregate productivity rises by 1.75%. Taking into account the transition path to a new steady state, the representative household's welfare rises by the equivalent of a 5.58% permanent increase in consumption. There are further welfare gains of 6.44% (permanent consumption equivalent) from eliminating the aggregate markup, which inefficiently lowers the labor supply. We keep labor supply fixed, so this effect is also absent from our model.

most establishments.

The right panel of Figure 6 depicts each firm's production per establishment, measure of additional establishments (beyond their initial unit measures), and total production in the first best relative to the market equilibrium. For production per establishment, the line is increasing because the planner reallocates production from small firms with small establishments to large firms with large establishments. For additional establishments, the line is 0 for single-establishment firms and the smallest multi-establishment firms, then steadily increases and flattens out below 1. This indicates that the planner does not open establishments at single-establishment firms and closes establishments at all multi-establishment firms—pushing the smallest into single-establishment status. In total, the planner closes 25.5% of additional establishments, which is 6.3% of all establishments. Finally, the net effect is that the planner reallocates production from the smallest and the largest firms toward medium-sized firms, which are large single-establishment firms. Thus, the planner does not reallocate production toward the highest markup firms.

Sources of misallocation. We decompose the 0.81% increase in consumption from implementing the first best into three sources of misallocation. Table 6 summarizes the results. First, we compute the misallocation of production within firms. Suppose the planner can only reallocate production labor across establishments within each firm, but cannot change each firm's measure of establishments or total production labor. This improves consumption by 0.06%. Second, we compute the misallocation of production across firms. Suppose the planner can reallocate production labor across all establishments, but cannot change each firm's measure of establishments. This improves consumption by 0.48%, which implies that the gain from reallocating production across firms is 0.42% of market equilibrium consumption (0.48% - 0.06%). Finally, we compute extensive margin misallocation, i.e., the gains from changing the set of establishments. This is the remaining increase in consumption that results from going to the first best, which is 0.44% of market equilibrium consumption (0.81% - 0.48%).

Intuitively, it is not surprising that the misallocation of production across firms is much bigger than the misallocation of production within firms because we saw in Figure 5 that there is more dispersion in the marginal social value of production across firms than within firms. Moreover, it is not surprising that there are nearly as big gains comes from changing the set of establishments as from reallocating production across establishments because we saw in Figure 5 that there is a huge gap between the marginal social value of opening estab-

Table 6: Sources of Misallocation

Production within firms	**Production across firms**	**Set of establishments**	**Total misallocation**
0.06%	0.42%	0.33%	0.81%

lishments and of producing at existing establishments.

5.2 Firm Size-Dependent Policy

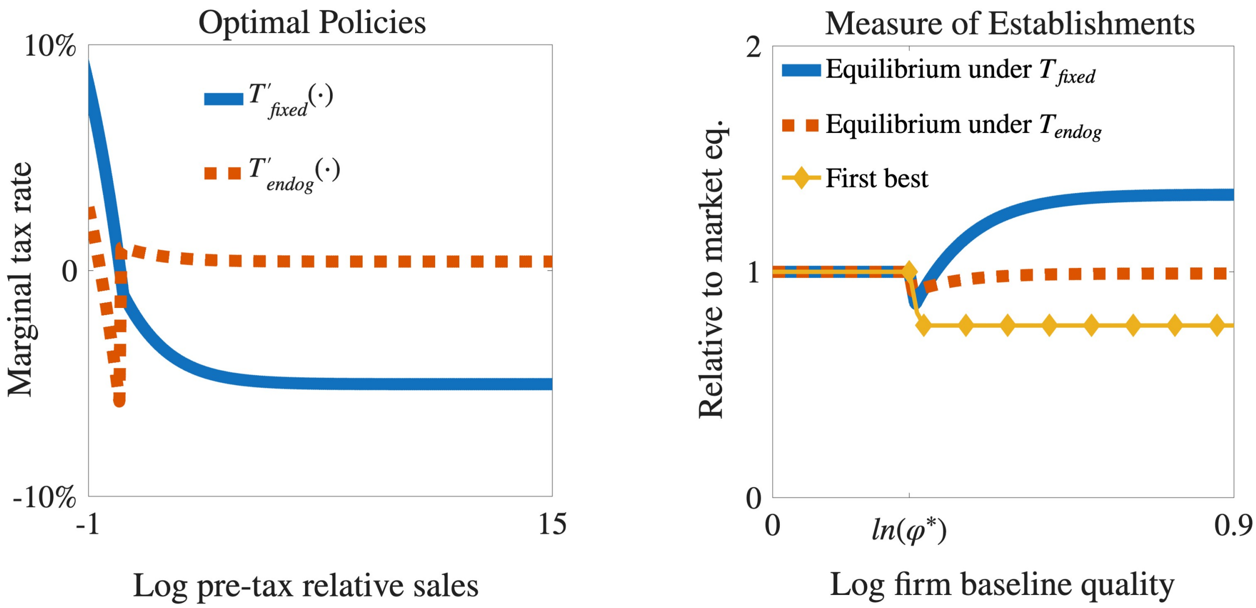
Finally, in our calibrated model, we study a firm size-dependent tax/subsidy $T(\cdot)$. If a firm has total nominal sales S across its establishments, then it pays $T(S)$, which is a tax if positive and a subsidy if negative. Hence, when choosing production at an establishment or how many establishments to open, its marginal revenue is multiplied by $1-T'(S)$. Any revenue (positive or negative) from the policy is given lump sum to the representative household. Without loss of generality, we restrict attention to revenue-neutral policies with $T(0) = 0$.[20]

First, we choose the policy $T(\cdot)$ to maximize equilibrium consumption while ignoring firms' extensive margin decisions. That is, hold fixed the set of establishments from the market equilibrium without policy. Then, the firm size-dependent policy $T(\cdot)$ is common knowledge and firms choose production at each establishment to maximize profits in a market equilibrium. The optimal policy, $T_{fixed}(\cdot)$, improves aggregate consumption by 0.41%. Recall that efficiently allocating production across the market equilibrium set of establishments increased consumption by 0.42%. Thus, the optimal policy undoes 98% of this misallocation. The left panel of Figure 7 illustrates the optimal marginal tax rate $T'_{fixed}(\cdot)$ as a function of firm sales. Larger firms face a lower marginal tax rate, so policy shifts production from small firms with small establishments to large firms with large establishments, undoing the misallocation implied by size-dependent establishment markups.

Now, is the policy $T_{fixed}(\cdot)$ still effective if the set of establishments responds endogenously in equilibrium? The answer is no. We need to take the extensive margin into account

[20]Revenue-neutrality means that in equilibrium, $\int_0^1 T(S(i))di = 0$. This is without loss of generality because if we multiply $1 - T'(\cdot)$ by $x > 0$, then the equilibrium wage also multiplies by x, so we decrease total revenue from the policy but do not affect firm incentives. $T(0)$ is also irrelevant because it simply shifts the household's total income between firm profits and lump sum transfers.

Figure 7: Optimal Firm Size-Dependent Policies



Left panel: a firm's marginal tax rates under optimal firm size-dependent policies as functions of its log pre-tax sales relative to aggregate output. Right panel: each firm's measure of establishments in the market equilibrium with each policy and in the first best, all relative to the market equilibrium without policy, as functions of the firm's log baseline quality.

when designing firm size-dependent policy. Formally, suppose the policy $T_{fixed}(\cdot)$ is common knowledge and then firms choose their measure of establishments and production at each establishment to maximize profits in a market equilibrium. Then, the policy backfires: consumption *falls* by 0.76% relative to no policy. To illustrate why, the right panel of Figure 7 depicts the equilibrium set of establishments under policy $T_{fixed}(\cdot)$ and in the first best relative to the market equilibrium without policy. Large firms respond to their relative subsidy under $T_{fixed}(\cdot)$ by opening many more establishments than is optimal in the first best. Thus, aggregate productivity rises by 2.6% relative to no policy, but at too high a cost: 2.9% of the labor supply shifts from production to opening establishments.

Intuitively, optimal policy under a fixed set of establishments backfires when the set of establishments is endogenously determined in equilibrium because there is substantial disagreement between how it is socially optimal to incentivize firms on the intensive margin (production at existing establishments) and on the extensive margin (opening/closing establishments). We saw this disagreement earlier in Figure 5, which showed that in the market equilibrium without policy, the marginal social value of opening establishments at a firm is well below the marginal social value of production even at much smaller firms. Hence, the optimal policy under a fixed set of establishments, $T_{fixed}(\cdot)$, gives too big a relative subsidy to large firms than is optimal under an endogenous set of establishments.

Finally, is firm size-dependent policy effective if we take into account that the set of establishments is endogenously determined in equilibrium? To answer this, suppose the policy is common knowledge and then firms choose their measure of establishments and production at each establishment to maximize profits in a market equilibrium. In this case, the optimal policy, $T_{endog}(\cdot)$, increases aggregate consumption by 0.06% relative to no policy. Thus, it undoes only 6% of the total losses from misallocation (0.05% out of 0.81%). Moreover, the gains from this policy are only 12% of the gains from $T_{fixed}(\cdot)$ when the set of establishments are held fixed. Hence, firms' extensive margin decisions sharply limit the effectiveness of firm size-dependent policy.

The limited effectiveness of firm size-dependent policy when the set of establishments is endogenously determined in equilibrium is again because there is disagreement between how it is socially optimal to incentivize firms on the intensive margin and on the extensive margin. The left panel of Figure 7 plots the optimal marginal tax rate $T'_{endog}(\cdot)$ and the right panel shows the resulting equilibrium set of establishments relative to the market equilibrium without policy. $T_{endog}(\cdot)$ actually taxes the largest firms with the highest markups because the dominant force is that it is socially optimal for them to close establishments. As a result of this tax, large firms close establishments, but far less than in the first best.

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A Additional Empirical Results

This section contains additional results related to our empirical findings. Figure 8 shows the size of successive establishments at firms with at least n establishments for $n \in \{3, 4, 5\}$, using employment and the wage bill to measure size. We see the same pattern as we saw for the wage bill among firms with at least 5 establishments: each successive establishment at a firm tends to be smaller, conditional on establishment age. The patterns are similar for $n > 5$ (not shown), but the confidence intervals become large and the results are less clear because the sample size becomes small; only 0.6% of our firm-year observations have more than 5 establishments.

Table 7 reports the first stage results of our instrumental variable regression listed in column 6 of Table 4 (from Section 2.4 on size-dependent markups). In the first stage, we predict a firm's log sales per establishment using the firm's previous year log sales per establishment, current log number of establishments, as well as fixed effects for the firm's 5-digit industry crossed with the year. This follows from the idea that a firm chooses its number of establishments ahead of time, but then its sales include noise.

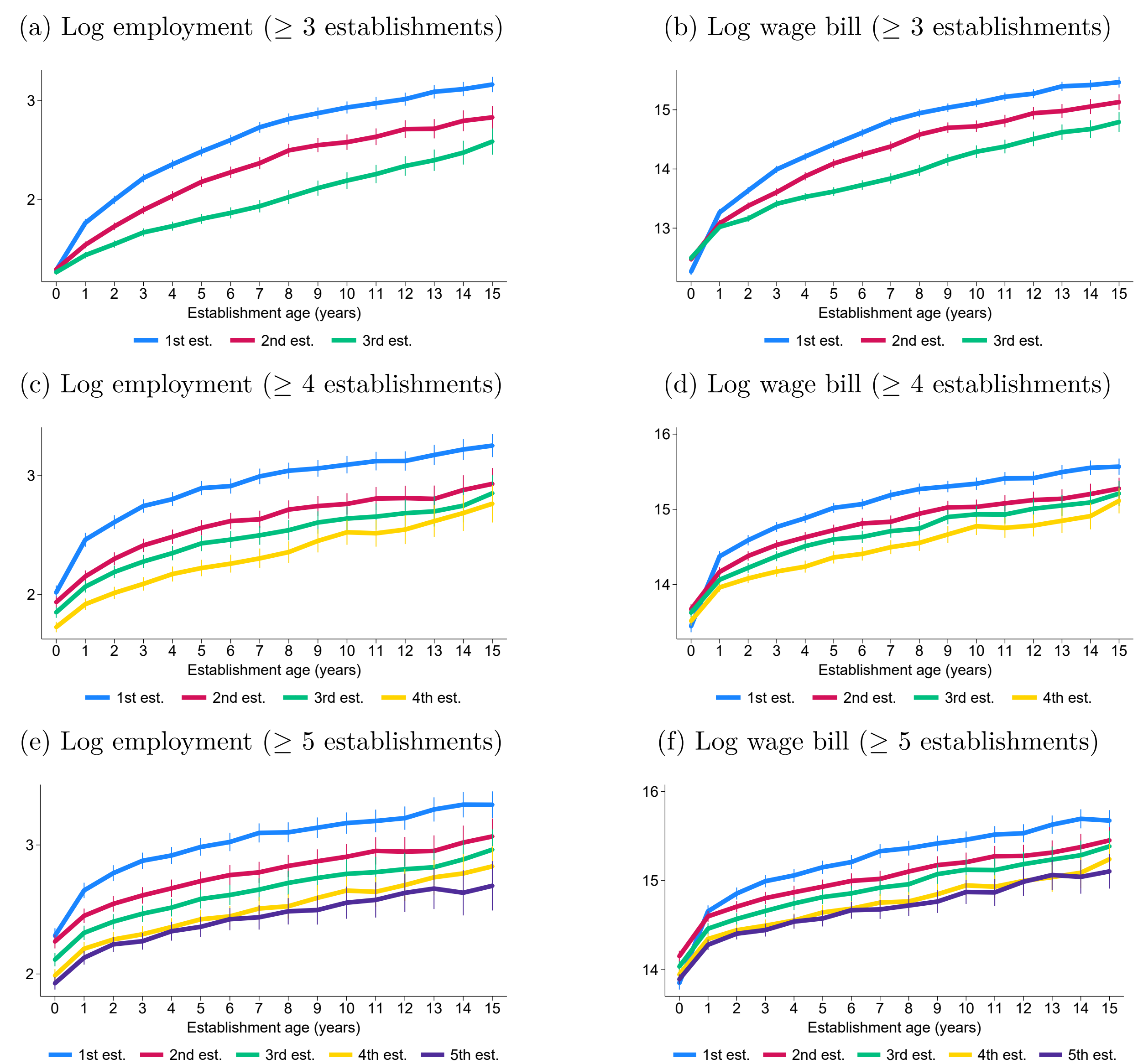
Table 7: Size-dependent markups: first stage IV

	Log sales per establishment$_t$
Log sales per establishment$_{t-1}$	0.878
	(0.006)
Log number of establishments$_t$	0.013
	(0.003)
Only multi-establishment firms	✓
Industry-year fixed effects	✓
R^2	0.860
Number of observations	86,515

Results from regressing a firm's log sales per establishment on the firm's previous year log sales per establishment, current log number of establishments, and fixed effects for the firm's 5-digit industry crossed with the year. We only use firm-year observations with more than one establishment. Standard errors (in parentheses) are clustered at the firm level.

The estimated coefficient on the lag of log sales per establishment is 0.878. Since this

Figure 8: Size of successive establishments



Average log employment (number of employees) and wage bill (in 2017 SEK) for firms' first $n \in \{3, 4, 5\}$ establishments as functions of establishment age, conditional on survival. Averages are computed across firms with at least n establishments. The vertical bars are 95% confidence intervals.

is close to 1, it indicates that sales per establishment is highly persistent and strongly autocorrelated, rather than being driven by random noise. The F-statistic testing the null hypothesis that lagged sales per establishment and current number of establishments are irrelevant is 11,712.54, meaning that the instruments are strong.

B Proofs

B.1 Proof of Theorem 1

Before we prove Theorem 1, it is useful to state and prove the following lemma.

Lemma 1. *Suppose the demand elasticity $\sigma(q)$ is strictly decreasing in relative output q and there exists a $\bar{q} > 0$ such that $\sigma(q) > 1$ for all $q \leq \bar{q}$. Then, for all $q \in (0, \bar{q})$,*

$$\frac{\Upsilon(q)}{q\Upsilon'(q)} < \frac{\sigma(q)}{\sigma(q) - 1} \tag{8}$$

and $\frac{\Upsilon(q)}{q\Upsilon'(q)}$ is strictly increasing in q.

Proof. We first show that inequality (8) holds weakly in the limit as q goes to 0. Then, we show that if (8) does not hold for some $q \in (0, \bar{q})$, then it cannot hold weakly as q goes to 0. This is a contradiction, so it proves that (8) must hold for all $q \in (0, \bar{q})$. Finally, we show that $\frac{\Upsilon(q)}{q\Upsilon'(q)}$ is strictly increasing in q.

First, we show that $\lim\limits_{q\to 0}\frac{\sigma(q)}{\sigma(q)-1}$ exists. By assumption, $\sigma(q)$ is strictly greater than 1 for all $q \in (0, \bar{q})$ and is strictly decreasing in q. Therefore, $\frac{\sigma(q)}{\sigma(q)-1}$ is strictly greater than 1 for all $q \in (0, \bar{q}]$ and is strictly *increasing* in q. It follows that as q goes to 0, $\frac{\sigma(q)}{\sigma(q)-1}$ is strictly falling and is bounded below by 1. Thus, $\lim\limits_{q\to 0}\frac{\sigma(q)}{\sigma(q)-1}$ exists.

Next, we show that $\lim\limits_{q\to 0}\frac{\Upsilon(q)}{q\Upsilon'(q)}$ exists and is less than or equal to $\lim\limits_{q\to 0}\frac{\sigma(q)}{\sigma(q)-1}$. The derivative of $q\Upsilon'(q)$ with respect to q is $\Upsilon'(q) + q\Upsilon''(q)$, which equals $\Upsilon'(q)\frac{\sigma(q)-1}{\sigma(q)}$. Since $\sigma(q)$ is strictly greater than 1 for all $q \in (0, \bar{q})$, it follows that $q\Upsilon'(q)$ is strictly increasing in q for all $q \in (0, \bar{q})$. Therefore, as q goes to 0, $q\Upsilon'(q)$ is strictly falling and bounded below by 0. Thus, $\lim\limits_{q\to 0} q\Upsilon'(q)$ exists and is a weakly positive real number. Now, there are two cases. For case 1, suppose $\lim\limits_{q\to 0} q\Upsilon'(q) > 0$. Then, since $\Upsilon(0) = 0$, it follows that $\lim\limits_{q\to 0}\frac{\Upsilon(q)}{q\Upsilon'(q)} = 0$. For case 2, suppose $\lim\limits_{q\to 0} q\Upsilon'(q) = 0$. Then, since $\Upsilon(0) = 0$, it follows that both the numerator and denominator of $\frac{\Upsilon(q)}{q\Upsilon'(q)}$ go to 0 as q goes to 0. To use L'Hôpital's rule, take the derivative of the numerator over the derivative of the denominator to get $\frac{\Upsilon'(q)}{\Upsilon'(q)+q\Upsilon''(q)}$, which equals $\frac{\sigma(q)}{\sigma(q)-1}$. Since $\lim\limits_{q\to 0}\frac{\sigma(q)}{\sigma(q)-1}$ exists, it then follows from L'Hôpital's rule that $\lim\limits_{q\to 0}\frac{\Upsilon(q)}{q\Upsilon'(q)} = \lim\limits_{q\to 0}\frac{\sigma(q)}{\sigma(q)-1}$. Hence, in either case, $\lim\limits_{q\to 0}\frac{\Upsilon(q)}{q\Upsilon'(q)}$ exists and is less than or equal to $\lim\limits_{q\to 0}\frac{\sigma(q)}{\sigma(q)-1}$.

Now, suppose $\frac{\Upsilon(q_0)}{q_0\Upsilon'(q_0)} \geq \frac{\sigma(q_0)}{\sigma(q_0)-1}$ for some $q_0 \in (0, \bar{q})$. Then, the derivative of $\frac{\Upsilon(q)}{q\Upsilon'(q)}$ with respect to q must be weakly negative at $q = q_0$ because

$$\frac{\partial \frac{\Upsilon(q)}{q\Upsilon'(q)}}{\partial q} = \frac{1}{q} - \frac{\Upsilon(q)(\Upsilon'(q) + q\Upsilon''(q))}{(q\Upsilon'(q))^2} = \frac{1}{q}\left(1 - \frac{\Upsilon(q)}{q\Upsilon'(q)}\frac{\sigma(q)-1}{\sigma(q)}\right). \tag{9}$$

It follows that if we decrease q below q_0, then $\frac{\Upsilon(q)}{q\Upsilon'(q)}$ weakly rises. Moreover, by assumption, $\frac{\sigma(q)-1}{\sigma(q)}$ is strictly positive at $q = q_0$ and strictly rises if we decrease q. Therefore, if we decrease q below q_0, then the derivative of $\frac{\Upsilon(q)}{q\Upsilon'(q)}$ with respect to q becomes strictly negative. This same argument shows that for all $q \in (0, q_0)$, the derivative of $\frac{\Upsilon(q)}{q\Upsilon'(q)}$ with respect to q is strictly negative. Moreover, by assumption, for all $q \in (0, q_0)$, $\frac{\sigma(q)}{\sigma(q)-1}$ is strictly increasing in q. Thus, since $\frac{\Upsilon(q_0)}{q_0\Upsilon'(q_0)} \geq \frac{\sigma(q_0)}{\sigma(q_0)-1}$, it follows that $\lim_{q\to 0}\frac{\Upsilon(q)}{q\Upsilon'(q)} > \lim_{q\to 0}\frac{\sigma(q)}{\sigma(q)-1}$. This is a contradiction because we saw earlier that the opposite inequality weakly holds. It follows that for all $q \in (0, \bar{q})$, inequality (8) holds strictly. Finally, from (9), it follows that for all $q \in (0, \bar{q})$, the derivative of $\frac{\Upsilon(q)}{q\Upsilon'(q)}$ with respect to q is strictly positive, so $\frac{\Upsilon(q)}{q\Upsilon'(q)}$ is strictly increasing in q. This completes the proof of the lemma. ■

We can now prove Theorem 1.

Proof. Suppose a planner has an infinitesimal quantity Δ of labor beyond the inelastic labor supply $\bar{L}$. First, suppose they use this labor to increase production at firm i's establishment n, holding fixed the measure of establishments at all firms from the market equilibrium, and holding fixed production at all other establishments from the market equilibrium. Then, household consumption (final good output) rises by $V_{prod}(i,n)\Delta$, where $V_{prod}(i,n)$ must be such that the Kimball aggregator constraint (3) continues to hold:

$$0 = \varphi(i,n)\Upsilon'(q(i,n))\frac{1}{Y}\Delta - \left(\int_0^1 \int_0^{N(j)} \varphi(j,m)\Upsilon'(q(j,m))q(j,m)\frac{1}{Y}\right) dmdjV_{prod}(i,n)\Delta,$$

where the first term on the right-hand side is the effect of the increase in relative output at firm i's establishment n and the second term is the effect of the resulting decrease in relative output at all other establishments due to the increase in final good output. It follows that

$$V_{prod}(i,n) = \varphi(i,n)\Upsilon'(q(i,n))D = p(i,n) = \frac{\sigma(q(i,n))}{\sigma(q(i,n))-1}W, \tag{10}$$

which follows from the derivation of the demand curve (4), where D is the demand index, and from expression (5) for an establishment's market equilibrium markup.

Next, suppose the planner uses the additional Δ units of labor to open establishments at firm i and produce the market equilibrium quantity, $y(i, N(i))$, at those establishments, holding fixed the measure of establishments at all other firms and production at all other establishments from the market equilibrium. The planner opens $\frac{\Delta}{\kappa(i,N(i))+y(i,N(i))}$ establishments at firm i because each establishment uses $\kappa(i, N(i))$ units of labor for the opening cost and $y(i, N(i))$ units of labor for production. Then, household consumption (final good output) must rise by $V_{estab}(i)\Delta$ where $V_{estab}(i)$ is such that the Kimball aggregator constraint (3) continues to hold:

$$0 = \frac{\varphi(i, N(i))\Upsilon(q(i, N(i)))\Delta}{\kappa(i, N(i)) + y(i, N(i))} - \left(\int_0^1 \int_0^{N(j)} \varphi(j, m)\Upsilon'(q(j, m))q(j, m)\frac{1}{Y}dmdj\right) V_{estab}(i)\Delta,$$

where the right-hand side of the first line is the effect of the increase in establishments at firm i and the second line is the effect of the resulting decrease in relative output at all other establishments due to the increase in final good output. It follows that

$$V_{estab}(i) = \frac{\varphi(i, N(i))\Upsilon(q(i, N(i)))Y}{\kappa(i, N(i)) + y(i, N(i))}D = \frac{p(i, N(i))y(i, N(i))}{W(\kappa(i, N(i)) + y(i, N(i)))}\frac{\Upsilon(q(i, N(i)))}{q(i, N(i))\Upsilon'(q(i, N(i)))}W,$$

which follows from the derivation of the demand curve (4), where D is the demand index. The first ratio after the second equal sign is revenue at firm i's marginal establishment relative to cost (including the establishment opening cost). If firm i is multi-establishment, then this ratio is 1 (otherwise the firm would open/close establishments), so

$$V_{estab}(i) = \frac{\Upsilon(q(i, N(i)))}{q(i, N(i))\Upsilon'(q(i, N(i)))}W. \tag{11}$$

On the other hand, if firm i is single-establishment, then

$$V_{estab}(i) = \frac{\sigma(i, 1)/(\sigma(i, 1) - 1)}{\kappa(i, 1)/y(i, 1) + 1}\frac{\Upsilon(q(i, 1))}{q(i, 1)\Upsilon'(q(i, 1))}W \leq \frac{\Upsilon(q(i, 1))}{q(i, 1)\Upsilon'(q(i, 1))}W, \tag{12}$$

where we use expression (5) for an establishment's market equilibrium markup and where the inequality follows because otherwise the firm would open establishments.

Now, suppose the demand elasticity $\sigma(q)$ is constant in an establishment's relative output q at $\sigma > 1$ (it must be greater than 1 by Assumption 1). It follows from (10) that for all firms i and establishments n, the marginal social value of production is constant at $V_{prod}(i, n) = W\sigma/(\sigma - 1)$. Moreover, since $\sigma = -\Upsilon'(q)/(q\Upsilon''(q))$ by definition, it follows that for all q, $\Upsilon'(q) = Aq^{-1/\sigma}$ for some $A > 0$. Along with $\Upsilon(0) = 0$, this implies that for all q, $\Upsilon(q) = A\frac{\sigma}{\sigma-1}q^{(\sigma-1)/\sigma}$. Thus, it follows from (11) that for all multi-establishment firms i, the marginal social value of opening establishments is constant at $V_{estab}(i) = W\sigma/(\sigma-1)$, which is equal to the constant value of all $V_{prod}(i,n)$. It follows from (12) that for all single-establishment firms i, $V_{estab}(i)$ is weakly lower. This completes the proof of the first statement of the theorem.

On the other hand, suppose the demand elasticity $\sigma(q)$ is strictly decreasing in an establishment's relative output q and is strictly greater than 1 for all $q < \bar{q}$ (the latter must be the case by Assumption 1). Since $\sigma(q)/(\sigma(q)-1)$ is strictly decreasing in $\sigma(q)$, it follows from (10) that $V_{prod}(i,n)$ is strictly increasing in the relative output of firm i's establishment n, $q(i,n)$. Moreover, it follows from (11) and Lemma 1 that if firm i is multi-establishment, then $V_{estab}(i)$ is strictly increasing in the relative output of firm i's marginal establishment, $q(i,N(i))$. Then using (12), it follows that if firm i is single-establishment, firm j is multi-establishment, and $q(i,N(i)) \leq q(j,N(j))$, then $V_{estab}(i) < V_{estab}(j)$. Finally, it follows from (10), (11), (12), and Lemma 1 that for all firms i, $V_{estab}(i) < V_{prod}(j,n)$ if $q(i,N(i)) \leq q(j,n)$. This completes the proof of the theorem. ■